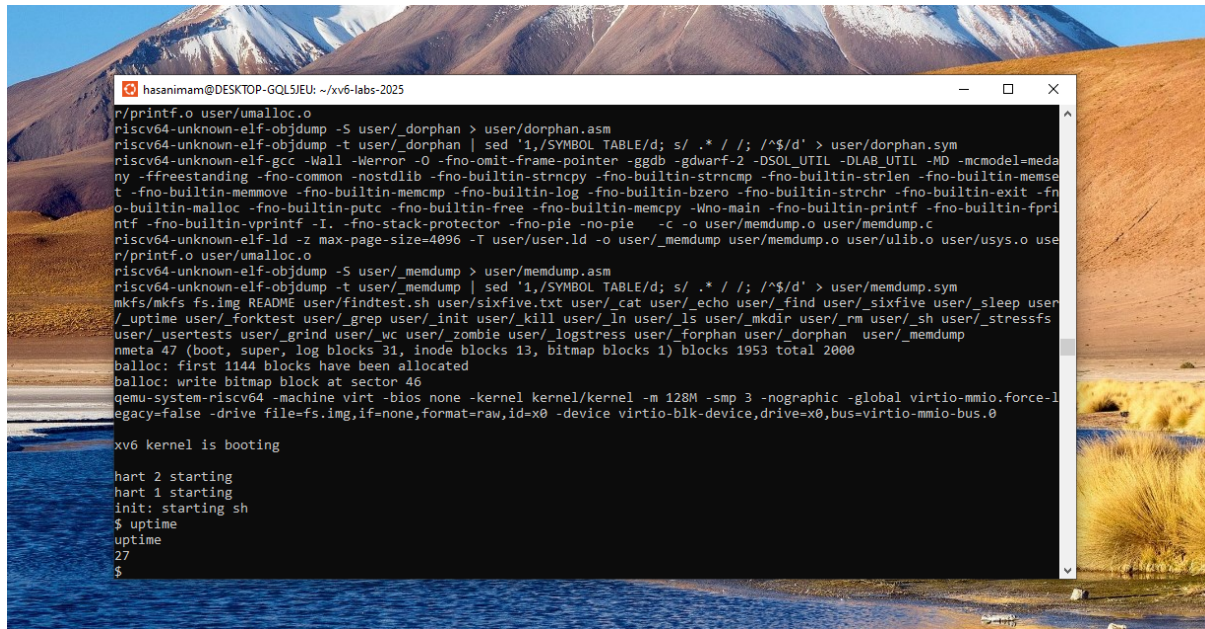


Syed Hasan Imam 29174

First Assignment (optional exercises)

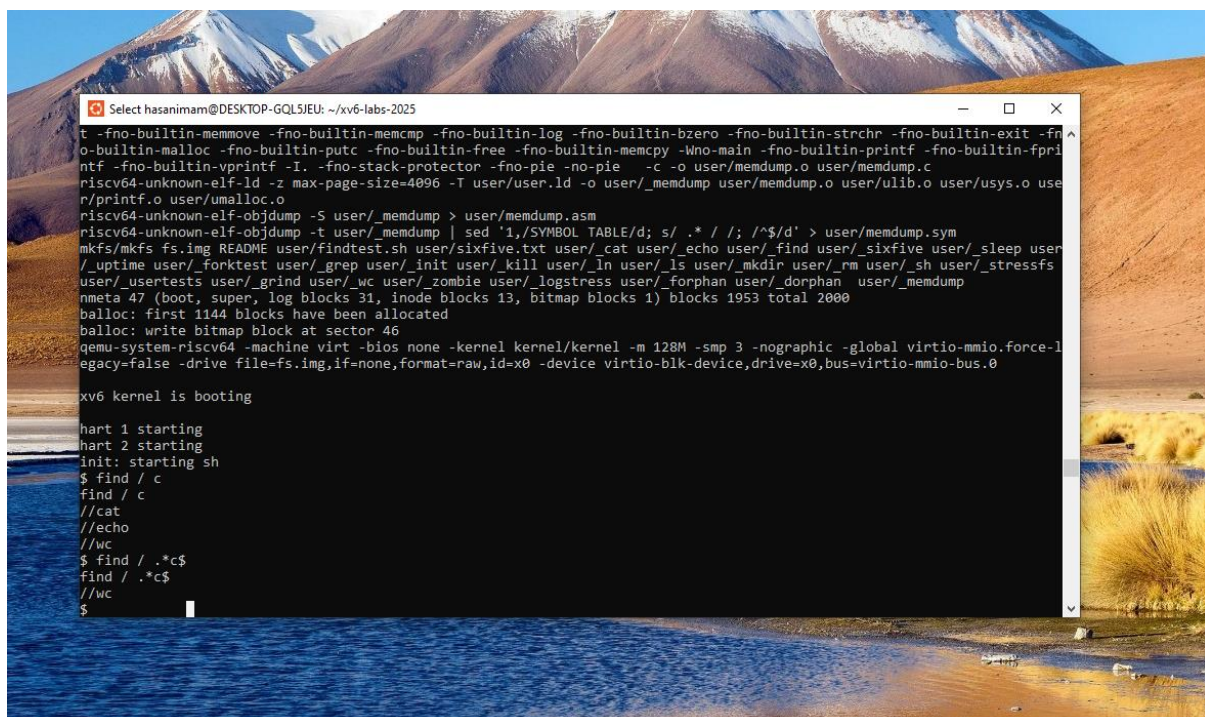
Q1. (uptime)



```
hasanimam@DESKTOP-GQL5IEU: ~/xv6-labs-2025
r/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_dorphan > user/dorphan.asm
riscv64-unknown-elf-objdump -t user/_dorphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/dorphan.sym
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_memdump > user/memdump.asm
riscv64-unknown-elf-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_find user/_sixfive user/_sleep user/_uptime user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertest5 user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1144 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ uptime
uptime
27
$
```

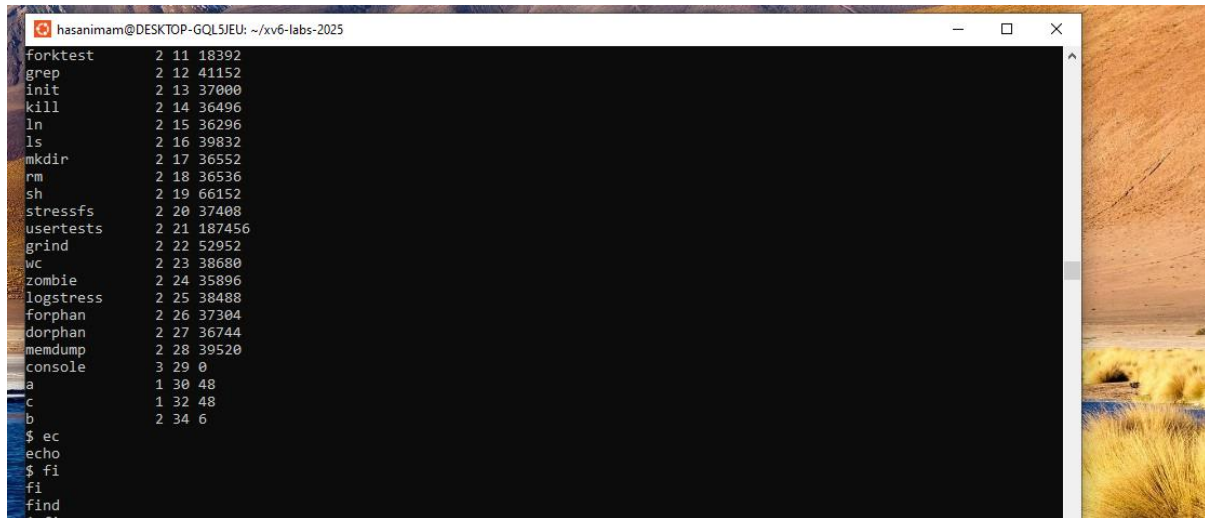
Q2.(find regex)



```
Select hasanimam@DESKTOP-GQL5IEU: ~/xv6-labs-2025
t -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_memdump > user/memdump.asm
riscv64-unknown-elf-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_find user/_sixfive user/_sleep user/_uptime user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertest5 user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1144 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ find / c
find / c
//cat
//echo
//wc
$ find / .*c$
find / .*c$
//wc
$
```

Q3.(tab completion)

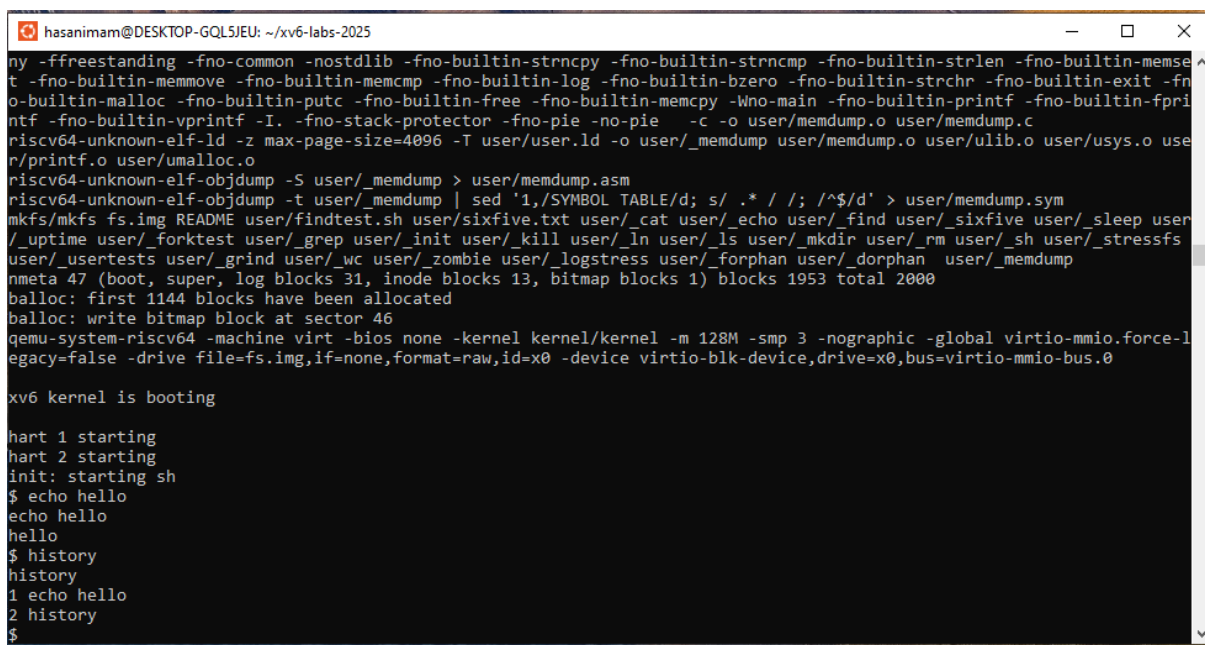


A terminal window titled 'hasanimam@DESKTOP-GQL5JEU: ~/xv6-labs-2025'. It displays a list of programs and their corresponding PID and PPID. The list is as follows:

Program	PID	PPID
forktest	2	11
grep	2	12
init	2	13
kill	2	14
ln	2	15
ls	2	16
mkdir	2	17
rm	2	18
sh	2	19
stressfs	2	20
usertests	2	21
grind	2	22
wc	2	23
zombie	2	24
logstress	2	25
forphan	2	26
dorphan	2	27
memdump	2	28
console	3	29
a	1	30
c	1	32
b	2	34

Below the list, the user enters '\$ ec' and 'echo', and the terminal shows 'echo' as the result of tab completion.

Q3(history)



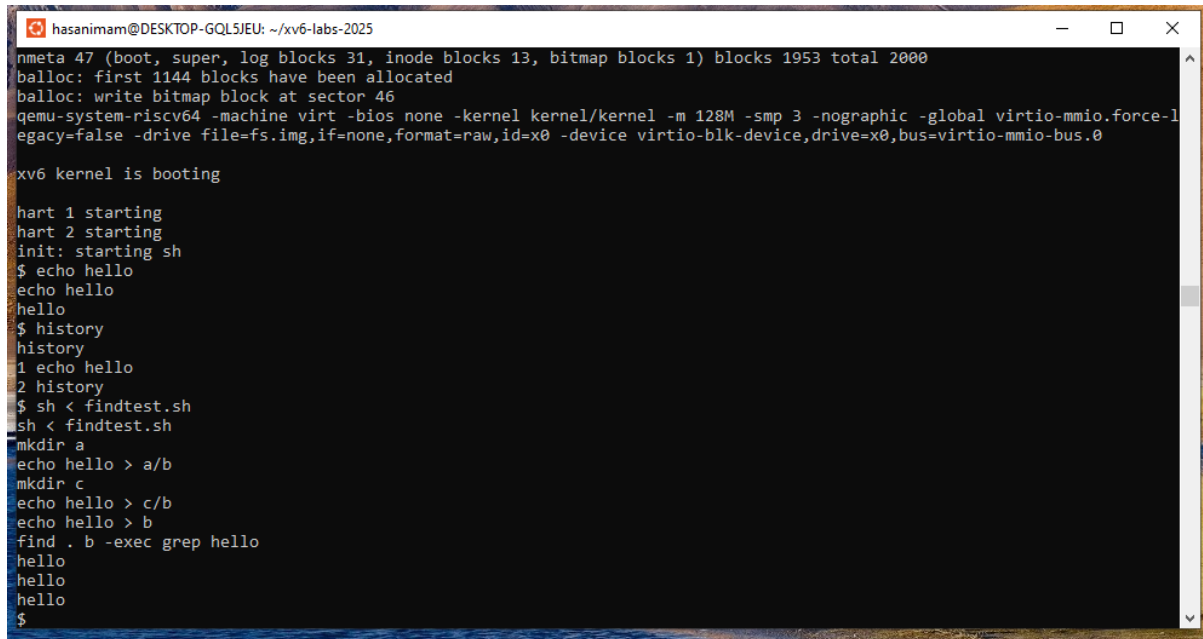
A terminal window titled 'hasanimam@DESKTOP-GQL5JEU: ~/xv6-labs-2025'. It shows the output of a QEMU system running xv6. The output includes the following commands and their results:

```
ny -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memse
t -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fn
o-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fpri
ntf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o use
r/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_memdump > user/memdump.asm
riscv64-unknown-elf-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_find user/_sixfive user/_sleep user
/_uptime user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs
user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1144 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-l
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo hello
echo hello
hello
$ history
history
1 echo hello
2 history
$
```

Q3(not print a \$ when processing shell commands)

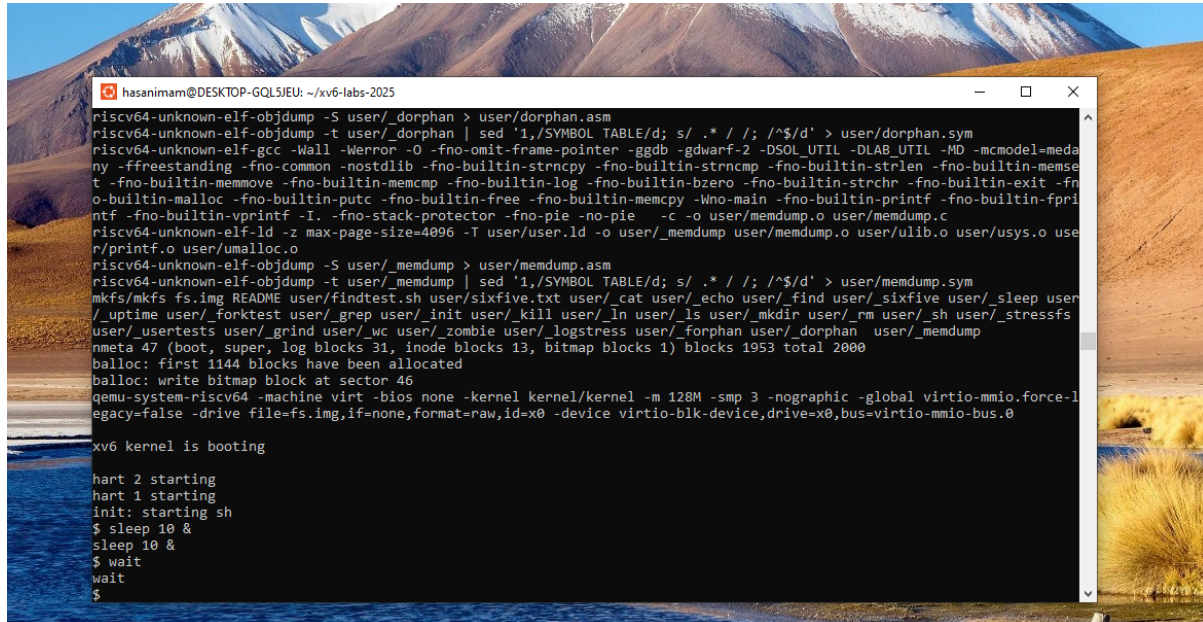


```
hasanimam@DESKTOP-GQL5JEU: ~/xv6-labs-2025
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1144 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo hello
echo hello
hello
$ history
history
1 echo hello
2 history
$ sh < findtest.sh
sh < findtest.sh
mkdir a
echo hello > a/b
mkdir c
echo hello > c/b
echo hello > b
find . b -exec grep hello
hello
hello
hello
$
```

Q3(wait)



```
hasanimam@DESKTOP-GQL5JEU: ~/xv6-labs-2025
riscv64-unknown-elf-objdump -S user/_dorphan > user/dorphan.asm
riscv64-unknown-elf-objdump -t user/_dorphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/dorphan.sym
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcpu=meda
ny -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memse
t -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fn
o-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fpri
ntf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o use
r/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_memdump > user/memdump.asm
riscv64-unknown-elf-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_find user/_sixfive user/_sleep user
/_uptime user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs
user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1144 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sleep 10 &
sleep 10 &
$ wait
wait
$
```

Problem set 1

Make grade:

```
hasaninam@DESKTOP-GQLS8U ~iv6-lab-2025
-- Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (0.7s)
-- Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: FAIL (1.3s)
...
$ echo abcdefgh12345678 | memdump ii
echo abcdefgh12345678 | memdump ii
0000 104023480
0000 175100885
$ echo abcdefgh12345678 | memdump S
echo abcdefgh12345678 | memdump S
0000 abcdefgh12345678
$ echo abcdefgh12345678 | memdump p
echo abcdefgh12345678 | memdump p
0000766056036261
$ qemu-system-riscv64: terminating on signal 15 from pid 11684 (make)
MESSAGE: "64636261"
QEMU output saved to xv6.out.memdump_examples
-- Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.0s)
-- Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.0s)
-- Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.3s)
-- Test exec ==
$ make qemu-gdb
exec: OK (0.8s)
-- Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: FAIL (1.0s)
Number of appearances of 'hello'
got:
6
expected:
5
QEMU output saved to xv6.out.find_exec_args
-- Test exec, recursive find ==
$ make qemu-gdb
(timeout) exec, recursive find: FAIL (30.3s)
Number of appearances of 'hello'
got:
7
expected:
3
QEMU output saved to xv6.out.test_find_sh
-- Test time ==
time: FAIL
Cannot read time.txt
Score: 180/131
make: *** [makefile:362: grade] Error 1
hasaninam@DESKTOP-GQLS8U ~iv6-lab-2025$
```

Activate Windows
Go to Settings to activate Windows.